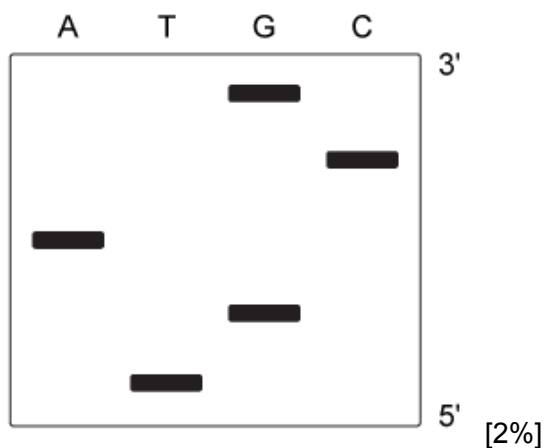


Researchers are screening bacterial chromosomes for mutations affecting protein function. After using DNA sequencing analysis and analyzing the gel electrophoresis results, they identify a sequence on the gel with a single point mutation in which the fourth nucleotide has been changed from adenine to cytosine. This sequence is shown below.

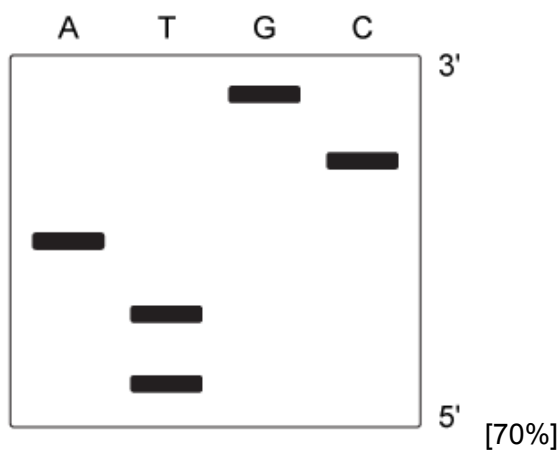
5'-TTACG-3'

Which of the following gel electrophoresis diagrams obtained from DNA sequencing analysis represents this sequence?

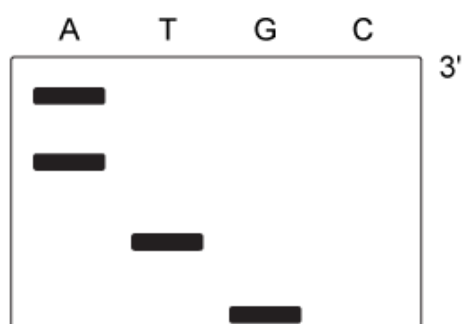
☐ A.

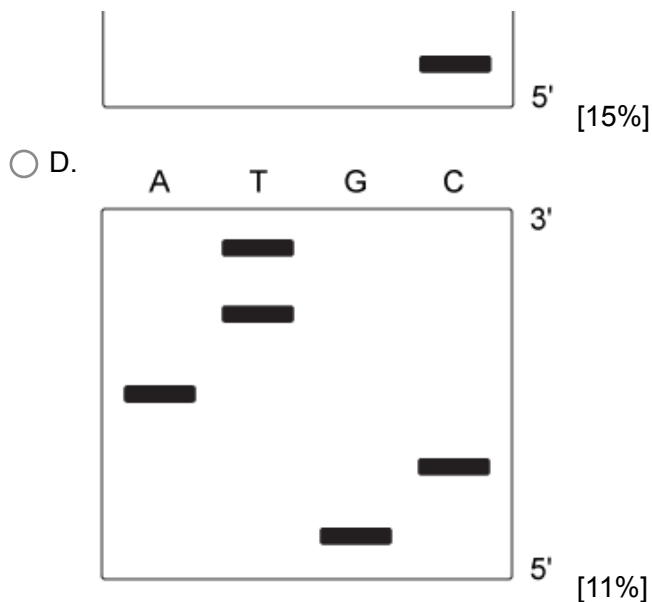


✓ ☒ B.



☐ C.

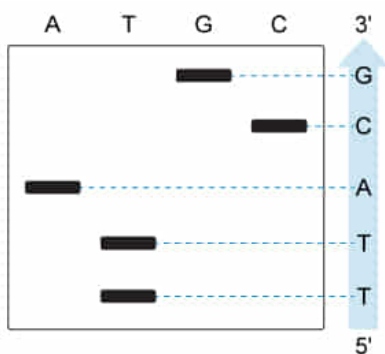




Correct

70% answered correctly

Explanation:



Reading the gel from 5' to 3' gives 5' – TTACG – 3'

Researchers use DNA sequence analysis to determine the specific nucleotide order of a DNA molecule. One method of DNA sequence analysis is the dideoxy (Sanger) sequencing method, which is carried out as follows:

1. The target DNA is isolated from cells or tissue and amplified using **PCR** or **gene cloning** to ensure that an adequate amount of DNA is available for the sequencing process.
2. The amplified DNA sample is split into four separate PCR tubes, each containing a primer complementary to the target DNA sequence, DNA polymerase, and the four deoxyribonucleotides (dNTPs). In addition, each tube contains one fluorescently labeled (or radiolabeled) dideoxynucleotide (ddNTP): ddATP, ddTTP, ddGTP, or ddCTP. Unlike dNTPs, ddNTPs lack a free hydroxyl group and are unable to form phosphodiester bonds with additional nucleotides, terminating DNA replication. Therefore, differential incorporation of ddNTPs (in the 5' to 3' direction) during PCR generates DNA fragments of varying lengths.
3. **Gel electrophoresis** is used to separate the fragments by size, and bands are visualized via fluorescence or radiolabeling. Because longer fragments migrate slower than shorter fragments in a gel, early incorporation of ddNTPs results in shorter fragments that migrate farther in the gel, and later incorporation of ddNTPs results in longer fragments that travel a shorter distance in the gel.
4. The bottom band in the gel represents the first nucleotide at the 5' end of the sequence, and the top band represents the last nucleotide at the 3' end of the sequence. Because DNA is *double-stranded*, both the target sequence and DNA sequence shown on the gel (ie, the target's *complement*) are

present in the organism.

In this scenario, researchers identify a mutated bacterial sequence (5'-TTACG-3') *on the gel* resulting from DNA sequencing analysis. Only the schematic in **Choice B** shows a gel correctly depicting this sequence.

(Choice A) This gel shows the DNA sequence of interest with an *additional point mutation* (ie, the second T has been changed to a G).

(Choice C) Because the sequence (5'-TTACG-3') was said to be identified *on the gel*, the correct gel should depict this sequence exactly, *not* the *complement*.

(Choice D) This gel depicts the sequence in the *wrong orientation* (ie, 3' to 5').

Educational objective:

The dideoxy (Sanger) method of DNA sequencing analysis uses PCR along with the incorporation of ddNTPs and gel electrophoresis to determine the nucleotide order of target DNA molecules.

Time spent:0 sec

Subject:

Biology

QID:402104

System:

Molecular Biology